Jack P. DeMarinis

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Education

University of Rhode Island, Kingston, RI — May 2026
M.S. Electrical Engineering (Accelerated B.S./M.S. Program)
GPA: 4.00 / 4.00

University of Rhode Island, Kingston, RI — May 2025
B.S. Computer Engineering
Minor: Mathematics
GPA: 3.90 / 4.00

Technical Skills

GenAI / LLM: OpenAI API, Gemini API, LangChain, agentic architectures, MCP servers, retrieval-augmented generation (RAG), embeddings, prompt engineering, local small language models

Languages: Python, C++, C, Bash, JavaScript, HTML/CSS, MIPS Assembly, LC-3

Backend / Systems: Linux, Docker, Google Cloud Console (GCP)

ML / AI: PyTorch, Hugging Face Transformers and Datasets

Tools & Design: Visual Studio Code, Unity, Fusion 360, AutoCAD, MATLAB, Multisim, Mathcad, VHDL, OpenMV IDE, Git/GitHub, ROS / ROS2

Experience

Graduate Generative AI Research Assistant — Sep 2025 – Present
University of Rhode Island, Kingston, RI
- Designed agentic, multi-step decision pipelines with tool calling and structured outputs to improve reliability across complex workflows.
- Built reusable MCP-style tool interfaces connecting LLMs to external services and internal utilities.
- Developed Linux and Docker workflows to support consistent builds and deployments across machines.
- Implemented and maintained a PostgreSQL database to support reliable storage and fast query access.

Computer Engineering Intern — January 2026 – Present
FireSale, Remote
- Built custom LLM and TTS software tools using the OpenAI API and ElevenLabs API.
- Created standalone full-stack prototypes to quickly develop and demonstrate new ideas.
- Communicated effectively in a remote workspace to keep projects documented and execution on track.

Undergraduate Generative AI Research Assistant — May 2024 – Aug 2025
University of Rhode Island, Kingstown, RI
- Built LLM features using RAG, improving factual grounding across evaluation prompts.
- Integrated LLM capabilities using OpenAI's API in combination with GCP, connecting to Gmail and Google Calendar and implementing Google OAuth for secure sign-in.
- Developed and shipped backend services supporting AI workflows handling hundreds of requests per day.
- Evaluated retrieval quality with a custom LLM-as-judge pipeline and synthetic gold-chunk datasets.

Computer Engineering Intern — May 2024 – Jan 2025
Electro Standards Laboratories, Cranston, RI
- Developed Python and C software on embedded Linux platforms, improving system functionality.
- Diagnosed and resolved electrical and software issues using debugging, instrumentation, and logging.
- Produced clear technical documentation to improve maintainability and team handoff.

Software Engineering Intern Jun 2023 – Dec 2023
IGT, West Greenwich, RI

- Resolved Linux system issues using low-level command-line debugging and root-cause analysis.
- Developed Bash, C, and C++ code for device-level integrations and internal tooling.
- Implemented OCR pipelines and validation checks to improve system accuracy and robustness.

IT Consultant Sep 2022 – May 2024
URI IT Service Desk, Kingston, RI

- Provided technical support and customer service to assist and guide users of URI technologies and tools.
- Troubleshot and resolved a broad range of computer issues in macOS and Windows environments.
- Utilized a standard IT ticketing system to document issues, communicate updates, and solve problems.

Selected Projects

AI Meeting Assistant

- Shipped a production full-stack app and deployment workflow on Railway with a PostgreSQL-backed data model for durable meeting artifacts.
- Designed an LLM workflow with schema-first JSON outputs and validation to standardize summaries and action items for downstream integrations.
- Built reliability into the pipeline (retries, idempotent writes, explicit failure states) to prevent partial processing and inconsistent data.
- Created reusable prompts and templates to make the workflow repeatable and simple to enhance.

Senior Capstone: Robotic Assembly & Inspection

- Built a modular automation workcell for through-hole PCBA assembly and inspection.
- Designed end effectors, fixtures, and feeder hardware in CAD, iterated via rapid prototyping.
- Integrated OpenMV vision for pose/orientation detection and UART/serial signaling to control code.
- Wrote documentation and handoff notes (design decisions, layout, troubleshooting).
- Presented the final system to a large audience, covering results and design tradeoffs.

LLM Robotics

- Built and simulated a mobile robot using ROS, integrating odometry and RGB-D camera data streams.
- Worked with SLAM-generated localization and mapping outputs to support downstream state awareness.
- Designed ROS nodes to preprocess sensor data and publish structured state representations for consumption by higher-level software.

Arduino Mister & Water Level Control System

- Developed an Arduino-based embedded system for environmental monitoring and remote misting control.
- Integrated DHT11 temperature/humidity sensing, analog water level measurement, Bluetooth communication, and OLED visualization.
- Built a mobile interface for real-time monitoring and misting control using Bluetooth.
- Designed and assembled a custom enclosure in CAD using 3D-printed parts for a fully working prototype.

Activities & Honors

Dean's List (every semester undergraduate and graduate)
Raymond M. Wright FastTrack Scholarship (2025–2026) graduate school
URI Club Wrestling Team
Westfield High School Varsity Soccer Captain